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ORIGINAL PAPER

GAN-based text line segmentation method for challenging handwritten documents

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Abstract
Text line segmentation (TLS) is an essential step of the end-to-end document analysis systems. The main purpose of this step is to extract the individual text lines of any handwritten documents with high accuracy. Handwritten and historical documents mostly contain touching and overlapping characters, heavy diacritics, footnotes and side notes added over the years. In this work, we present a new TLS method based on generative adversarial networks (GAN). TLS problem is tackled as an image-to-image translation problem and the GAN model was trained to learn the spatial information between the individual text lines and their corresponding masks including the text lines. To evaluate the segmentation performance of the proposed GAN model, two challenging datasets, VML-AHTE and VML-MOC, were used. According to the qualitative and quantitative results, the proposed GAN model achieved the best segmentation accuracy on the VML-MOC dataset and showed competitive performance on the VML-AHTE dataset.

Keywords Text line segmentation · Generative adversarial networks · Document analysis · Handwritten document

1 Introduction

Historical handwritten documents are one of the most important sources for researchers in many fields such as literature, history, sociology, architecture, etc. To this end, governments, libraries, and collections have been digitizing handwritten documents accumulated over centuries to make them available to researchers. At this point, the use of document analysis methods becomes crucial for researchers to access information in the fastest and most accurate way.

Text line segmentation (TLS) is crucial for high-level document analysis tasks, such as word segmentation, word spotting, and text recognition. The TLS step needs to be overcome powerfully, as segmentation errors will lead to cumulative errors in subsequent steps. However, text line segmentation is still considered a challenging problem due to the difficulties inherent in the manuscript. Segmentation accuracy is degraded in cases such as skewed and curved lines, multi-directional lines, overlapping or touching each other of words or characters, different writing styles of different authors, etc.

TLS is a multilingual problem and the issues to be addressed are often common. Moreover, due to the existence of sidenotes and footnotes added over the years, the TLS problem can be more challenging in Arabic and Ottoman manuscripts. The challenges in these manuscripts are:

- Most of the documents include skewed and curved text lines. So, TLS methods should adapt to angular changes in any individual text line.
- The text lines in Arabic and Ottoman documents pick up diacritics and punctuation placed between consecutive text lines. So, it is important to assign these diacritics and punctuation accurately for further steps.
- The consecutive text lines may include touching and/or overlapping characters. So, in cases of touching or over-

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lapping, it is necessary to correctly identify and extract the real characters.
- The sidenotes and footnotes were written very near to the main body with different orientations. This leads to a sharp angular transition in a small text region and makes text line segmentation harder.

As stated above, TLS is a challenging problem in the case of documents with multi-skewed, and multi-directed text lines, documents with curved and fluctuated text lines, and documents with touching and overlapping characters or words. To overcome these challenges, in this paper, we propose a GAN-based TLS method. The proposed GAN model was trained with specially selected data sets, VML-AHTE and VML-MOC, to overcome these problems. To evaluate the performance of the proposed method and compare it with state-of-the-art methods same training and test data sets were used. TLS has been considered an image-to-image translation task. The GAN model has been trained for accurate extraction of each text line by generating segmentation masks. To the best of our knowledge, the proposed method is the first GAN model that generates masks in challenging documents. The main contributions of this work can be summarized as follows:

- A generative TLS model is proposed for the segmentation of handwritten documents including various challenging text line behaviors.
- The Segmentation performance of the model was evaluated on two different challenging datasets.
- The proposed model has the ability to segment the manuscripts including touching/overlapping characters, multi-directional text lines, and text lines with punctuation marks.

The rest of the paper is organized as follows: In Sect. 2, the related works are given. In Sect. 3 the proposed method is presented rigorously whereas Sect. 4 gives the qualitative and quantitative results. Finally, the results are discussed and a conclusion about the segmentation performance of the proposed method is given in the last section.

2 Related works

2.1 Learning-free methods for TLS

Various learning-free methods have been proposed for the TLS problem so far. Projection-based methods [1, 2, 32], Hough transform-based methods [20–22], and connected component-based methods [16, 30] are examples of learning-free methods frequently used in the literature. In projection-based methods, it's assumed that the text line is straight and segmentation is done by using the spaces between adjacent text lines. While projection-based methods are often successful in printed documents [2], they fail especially in manuscript documents containing skewed and curved text lines. Successful text line segmentation can be done up to a certain level by using piece-wise projections in handwritten documents that do not have multi-skewed text lines and do not contain too much complexity [1]. However, the range of document images that projection-based methods can successfully segment text lines is limited. Moreover,projection-based methods cannot perform successful segmentation due to the situations that make segmentation in handwritten documents difficult. Hough transform-based methods assume that the text line components are flat or with a low angle of inclination. For this reason, Hough transform-based methods can perform successful text line segmentation in printed document images, but not for handwritten document images. In Hough transform-based methods, especially in languages such as Arabic, where small components used for punctuation and emphasis are dense, the segmentation performance is considerably reduced in case of curvature between text lines. In connected component-based methods, each pixel is usually of equal importance. Therefore, connected component-based methods fail especially if the manuscript has touching and overlapping text lines. The above-mentioned learning-free methods cannot correctly segment text line documents with vertical, multi-skewed, and multi-directed text lines, documents with curved and fluctuated text lines, and documents with touching and overlapping characters or words. Until the last decade, the text line segmentation problem was usually tried to be solved with learning-free methods. However, to get rid of the constraints of learning-free methods, with the increase in computer processing capacities and data amount, machine learning and CNN-based text line segmentation methods have been used more frequently recently.

2.2 CNN-based methods for TLS

As learning-based approaches, handwritten text line segmentation methods using a fully convolutional network are proposed [3, 4, 24, 28, 33]. In [27], baseline detection on historical documents is performed using a convolutional neural network-based method. In another learning-based approach, text lines are segmented as X-height using adaptive U-net [23]. Fully convolutional neural network variants U-net [29] and RU-net are used for Arabic handwritten document segmentation [26]. In [12], text lines are extracted as baselines with ARU-Net based on U-net architecture. In [8], Mask R-CNN is used as a holistic method for text-line extraction in handwritten documents. The segmentation performance of the proposed method is evaluated by implementing various datasets including different kinds of challenges such as touching, overlapping characters, and diacritics. In [13], the

TLS problem was handled as an instance segmentation problem and implemented the model to various datasets.

2.3 GANs for TLS and document analysis

Generative adversarial networks (GAN) [6], which is one of the deep learning methods used as an effective method to generate images recently, have also been used to segment text lines in document images [7, 17]. In [17], the text line segmentation performance of the proposed GAN model is evaluated for different combinations of the GAN loss, L1 loss, and L2 loss functions. The GAN model is applied on two different datasets consisting of manuscript document images and the best text line segmentation performance is achieved with the GAN model in which U-Net [29] is used as a generator and PatchGAN [14] is used as a discriminator. In our previous work, text line segmentation is performed on a dataset consisting of Arabic manuscripts with compelling features in terms of segmentation with GAN [7]. Another important application of GAN is the segmentation of texts from touching text and non-text components [25]. For this purpose, a GAN model with ResNet as a generator and PatchGan as a discriminator is proposed. The results show that the proposed GAN model can efficiently predict the text regions from the combined (text and non-text) images.

3 Method

As already mentioned, handwritten documents have several challenging problems for accurate text line segmentation (TLS). Among these challenges, the presence of touching and overlapping characters, and skewed and multi-directed lines are the most bottleneck problems. In this work, we focused on producing a model that can overcome these problems in text line segmentation. For this purpose, the TLS problem was considered as an image-to-image translation problem and the proposed model was trained to generate the optimal mask for each text line. Here, the mask corresponds to a planar structure covering all characters and punctuation marks in an individual text line. The main purpose of the image-to-image translation is to transfer images from one domain to another by performing operations such as color space, histogram transformations, style, or characteristic transfer. Hereby, in this work, image-to-image translation was utilized to map the spatial information from the original image domain to the mask image domain and so generate a mask for each text line. A generative adversarial network (GAN) is one of the powerful CNN-based models for image-to-image translation tasks. In this work, we've been inspired by the pix2pix model [14]. pix2pix model is composed of two competitive CNN models, U-Net and PatchGAN. U-Net is used in the generator part of the model, and PatchGAN is used in the discriminator part. These models play a min–max game, whereby the generator aims to produce better synthetic images and the discriminator updates itself to better discriminate the synthetic images produced by the generator.

Text lines in document images with challenging segmentation characteristics were segmented using a generative adversarial network. The block diagram of the proposed method is given in Fig. 1. In the training phase of the network, original document images and ground truth images containing text line masks were used. The GAN network model was trained to learn the spatial relationship between the original manuscript texts and the mask images of the text lines. Training and test data were created based on a data set consisting of Arabic texts with challenging characteristics. Moreover, in order to examine the effect of loss functions on the text line segmentation, the training process of the network was carried out using three different loss functions, L1, L2, and GAN loss function.

3.1 Dataset

Since GAN and CNN-based models are highly data-dependent, the content and diversity of the dataset for accurate model training are crucial. From this viewpoint, a dataset consisting of two datasets with different text line characteristics was generated. The whole dataset has many samples involving challenges in text line segmentation, which is the motivation for this work. Training and test data were generated from the whole dataset. Apart from the test data, to evaluate the overall segmentation performance of the proposed GAN model, a separate test dataset was utilized. The characteristics of the datasets used and the processing of the data for the training and the evaluation of the GAN model are explained below.

3.1.1 VML-AHTE dataset: Arabic handwritten text line extraction dataset

VML-AHTE dataset [5] has characters that touch and overlap between adjacent text lines and too many diacritics. Out of a total of 30 pp. of high-resolution images in the dataset, 20 of them are set for training and 10 for testing. An example of a document image in the dataset and its corresponding ground truth image are given in Fig. 2.

3.1.2 VML-MOC: multiply oriented and curved handwritten text line dataset

The text lines in this dataset containing Arabic manuscripts, containing side notes over the years, are highly skewed and curved. In the VML-MOC dataset [18], there are 20 images for training and 10 for testing. In this dataset, text lines in ground truth images are represented by masks covering the

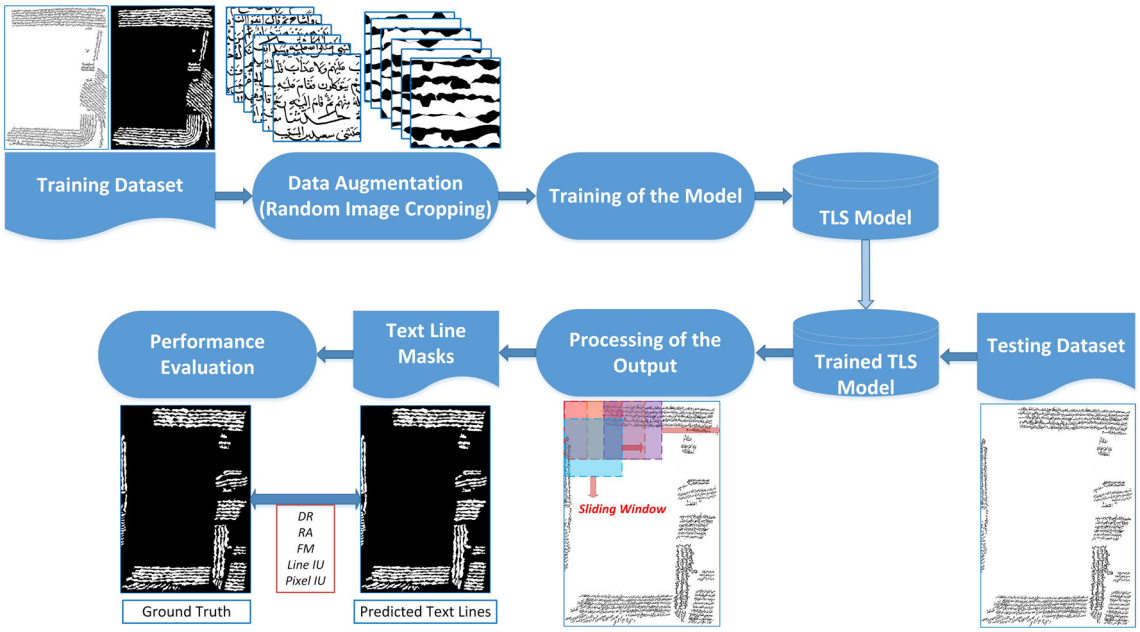


Fig. 1 The block diagram of the proposed method

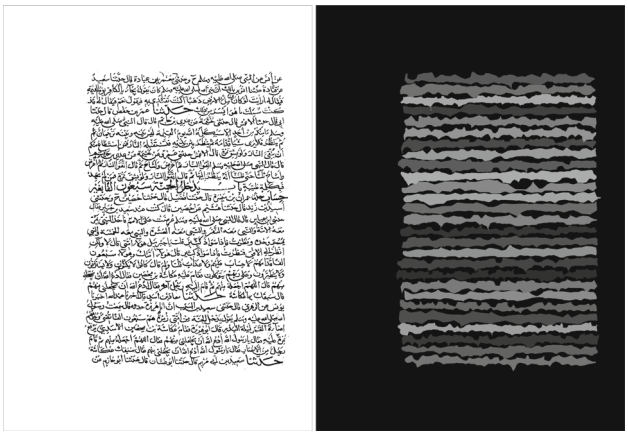


Fig. 2 An example image from the VML-AHTE dataset and its ground truth

text lines. An example of a document image in the dataset and its corresponding ground truth image are given in Fig. 3.

3.1.3 Preparing the dataset for training the network

In the VML-AHTE and VML-MOC datasets, there are a total of 60 high-resolution document images, 30 of each. The training and test data were generated from training and test images. For each training document image, the text density in the document image is calculated and a total of 50,000 training data of 512 × 512 size is created by calculating the text density in the document image and making sure that the content of the image cropped from random regions in direct proportion to this density is above a threshold value we set as 0 pixel. Our aim is to exclude image patches that do not contain any textual components from the training set. Examples from the training set are given in Fig. 4.

3.2 Evaluation

The pix2pix-based GAN Model designed for text line segmentation is given in Fig. 5. In the proposed method, the U-Net network model [29], which is also used in the segmentation of medical images as a generative network, is employed. The U-Net architecture consists of two parts: the encoder and the decoder. In the encoder part, the input image is transformed into feature maps using convolution and max-pooling layers. Each convolutional layer learns specific features of the image and produces a lower-resolution feature map, providing information about the general structure of the image. The transpose convolution layers in the decoder part rescale the feature maps, effectively increasing the size and bringing back the details in the image to generate the target image. Each downsampling block contains a 3 × 3 convolution, batch normalization, and a leaky ReLU layer with a slope of 0.2. In the upsampling blocks, there is a transposed convolution, Batch Normalization, a Dropout layer with a 50% rate, and a leaky ReLU layer with a slope of 0.2. Moreover, it includes skip connections, which associate the feature maps from the encoder part with higher-resolution output, allowing the network to learn better features and produce better results. In this work, the generative network takes original document images as input and produces text line masks as output. The number of convolution or transpose convolution layers used in the encoder and decoder parts

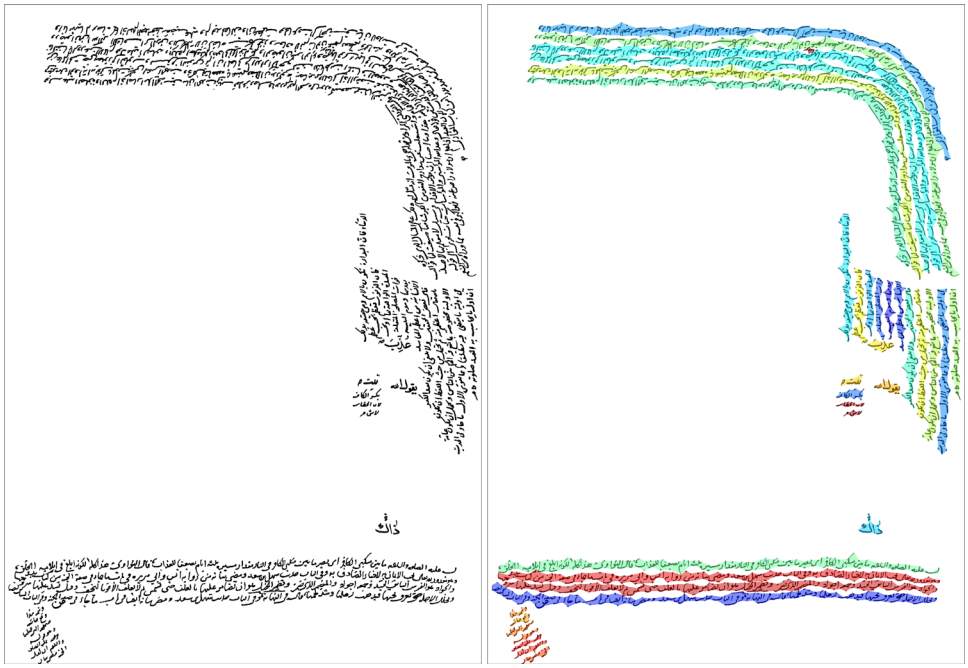


Fig. 3 An example document image and ground truth image from the VML-MOC dataset

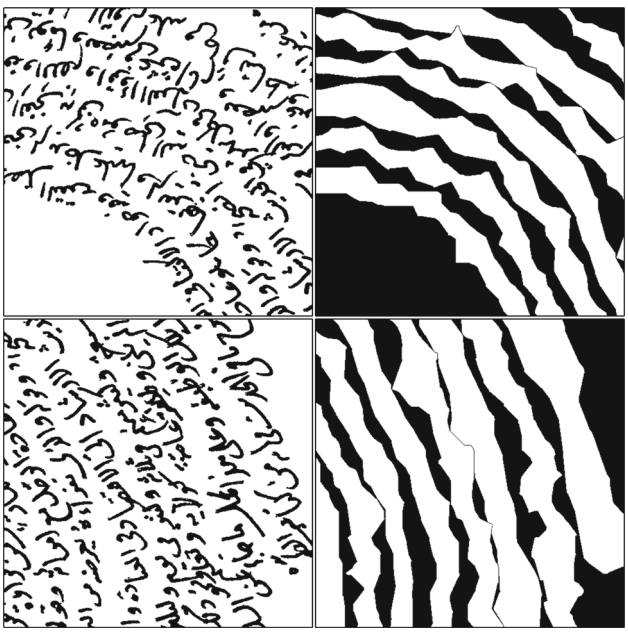


Fig. 4 Examples from the training set

can vary depending on the dataset or application, which also changes the depth of the network. In this work, the encoder and decoder depth of the U-Net network is set to 8, while the depth of PatchGan used as a discriminator is set to 4.

The model was trained for 30 epochs using 1 batch per iteration on NVIDIA RTX 8000 with 50,000 pieces of data generated. The training process took about 4 days for 30 epochs. In this work, mini-batch Stochastic Gradient Descent (SGD) was used during the training process of the network. Additionally, the Adam solver [15] was used with a learning rate of 0.0002, and momentum parameters were set to $\beta_1 = 0.5, \beta_2 = 0.999$. Initially, the objective function of the pix2pix model is expressed as a min-max game between generator(G) and discriminator (D), where the generator aims to minimize the objective function and the discriminator the opposite. As proven in [14, 17], L1 and L2 losses can outperform the original GAN loss function and produce better-quality masks for the text lines. In this work, L1 and L2 losses, given in Eqs. 1 and 2, are also applied and the segmentation performance of the method is examined for different losses.

$$L_{L1}(G) = E_{x,y,z}[\|(y - G(x,z))\|_1] \tag{1}$$

$$L_{L2}(G) = E_{x,y,z}[\|(y - G(x,z))\|_2] \tag{2}$$

The test images were allocated to determine the text line segmentation performance of the trained network. These test images were divided into non-overlapping 512×512-sized patches. By examining the pixel density of the divided patches, the text line estimation of the patches with the text was made with the GAN model, and the text line mask of that patch was obtained. In order to reproduce the segmented document image from the patches, problems arise due to the different mask estimations in the areas remaining on the patch edges when the obtained pieces are directly combined (Fig. 6a). To solve this problem, a sliding window with the size of 512×512 was used while the test images were passed through the GAN model. A 512×512 sliding window is applied to the test image and given as input to the model. A 300×300 model estimate at the center of the model output is

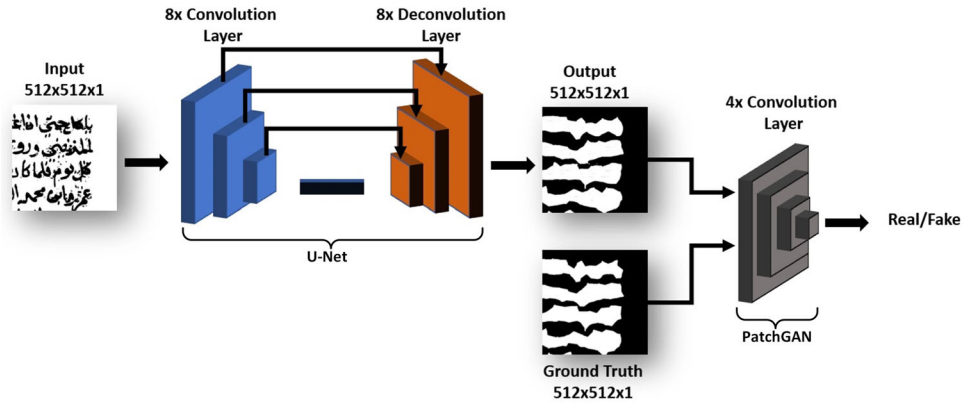


Fig. 5 Text line segmentation network

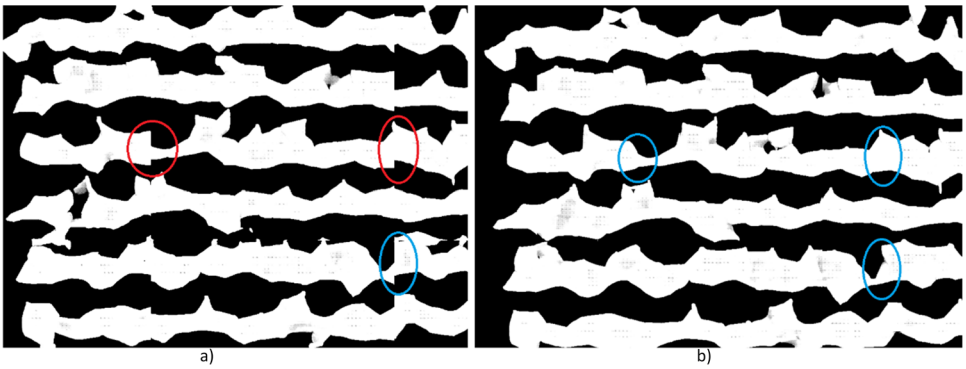


Fig. 6 Segmentation results. **a** without sliding window application **b** with sliding window application

taken and stored in a matrix. The horizontal and vertical step size of the sliding window is set as 300 pixels and this process is continued until all pixels in the image are processed. Thus, the non-overlapping regions of 300 pixels each are merged into the empty matrix, and the model is estimated against the original image. During the process, the text lines on the whole page were estimated by adding 300 × 300 sections in the middle of the document image at each window transition. As a result of this process, a more natural transition is obtained as can be seen in Fig. 6.

In order to evaluate the results and compare the segmentation performance with the state-of-the-art methods rigorously, we utilized several performance metrics. The first one is Intersection over Union (IU), which is evaluated on both pixel and line levels [8, 31]. Pixel IU is a measure of the ratio of correctly classified pixels to misclassified pixels and is calculated with:

$$PixelIU = \frac{\Sigma_{TP}}{\Sigma_{TP} + \Sigma_{FP} + \Sigma_{FN}} \tag{3}$$

Line IU is a measure of segmentation performance at line level. Line IU is calculated based on line precision and line recall metrics obtained with respect to a particular threshold value.

$$LinePrecision = \frac{LineTP}{LineTP + LineFP} \tag{4}$$

$$LineRecall = \frac{LineTP}{LineTP + LineFN} \tag{5}$$

$$LineIU = \frac{CL}{CL + ML + EL} \tag{6}$$

where CL is the number of correctly segmented text lines and a line is of CL if the line precision and the line recall are above the threshold value T. ML is the number of under-segmented text lines and a line feeds the ML value if the line recall is below the threshold value T. EL is the number of over-segmented text lines and when the line precision is below the threshold value T, the EL value increases. Here, the threshold value was set to 0.75 to compare our results with the ones of state-of-the-art methods.

The second performance metrics set is introduced in [11]. These metrics are the detection rate (DR), the recognition accuracy (RA), and the F-measure (FM) and are calculated depending on the match score obtained at a threshold value. The match score is calculated as:

$$MatchScore(i, j) = \frac{T|G_j \cap P_i|}{T|G_j \cup P_i|} \tag{7}$$

where G_j is the set of the pixels in j_{th} text line component of ground truth and P_i is the set of pixels in i_{th} component of the predicted mask. For a given pair (i, j) if the match score is equal to or above the threshold value, it represents a one-to-one match. Here, the threshold value was set to 0.9. Let

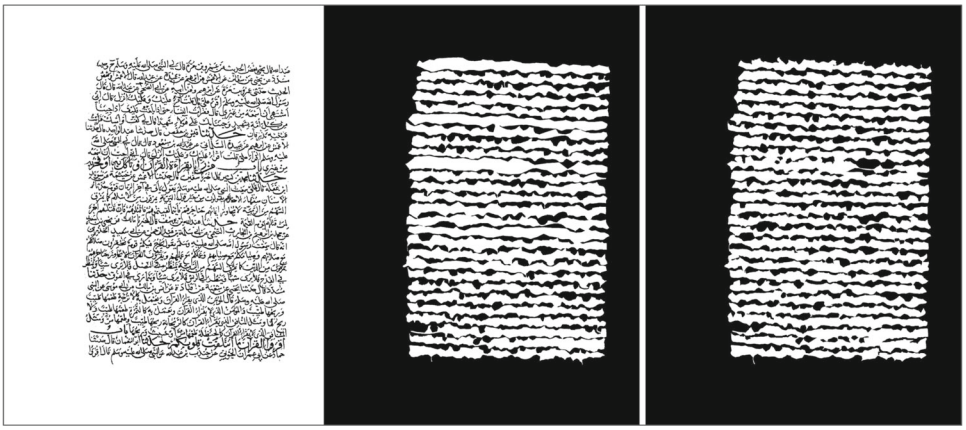


Fig. 7 VML-AHTE results: test image (left), ground truth (middle), GAN result (right)

Table 1 Performance metric values on the VML-AHTE dataset

Method	Learning	DR	RA	FM	Line IU	Pixel IU
Barakat et al. [9]	Unsupervised	–	–	–	99.28	91.40
UTLS [5]	Unsupervised	–	–	–	98.55	88.95
Barakat et al. [19]	Unsupervised	–	–	–	90.94	83.40
Mask-RCNN [8]	Supervised	84.43	58.89	68.77	93.08	86.97
Mask-RCNN + [8]	Supervised	88.14	87.78	87.96	97.83	89.61
FCN + EM [4]	Supervised	95.55	92.80	94.30	94.52	90.01
Islam et al. [13]	Supervised	–	–	–	97.10	96.60
GAN (GANLoss)	Supervised	83.78	79.01	81.26	99.01	95.03
GAN (L1)	Supervised	90.42	89.27	89.80	93.99	94.54
GAN (L2)	Supervised	87.65	88.02	87.73	89.99	93.29

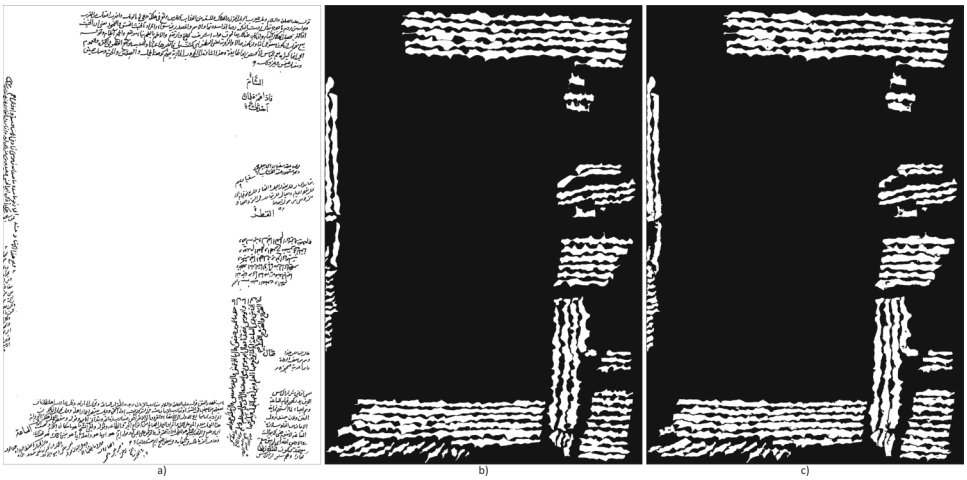


Fig. 8 **a** VML-MOC image **b** ground truth **c** segmentation result

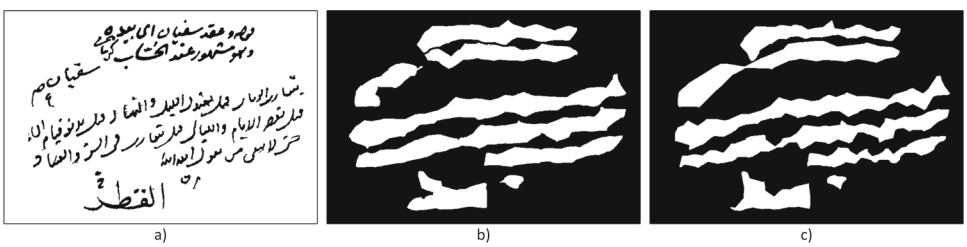


Fig. 9 **a** VML-MOC fragment image, **b** ground truth, **c** segmentation result

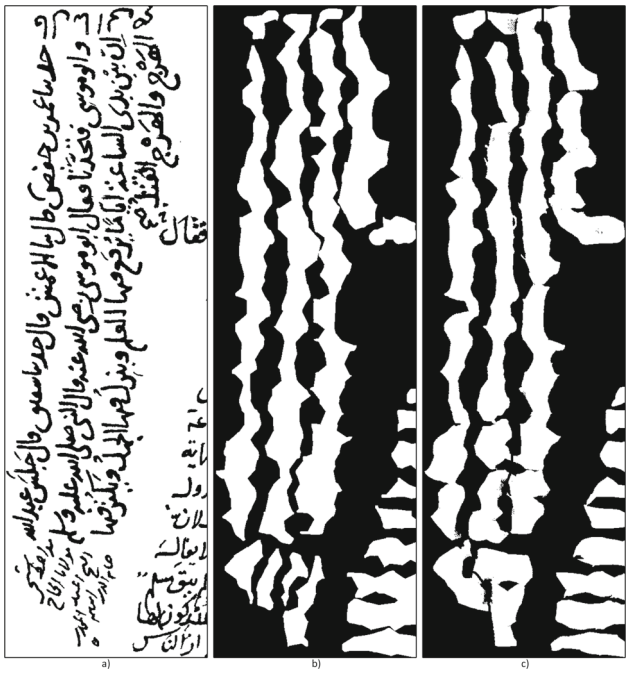


Fig. 10 **a** VML-MOC fragment image, **b** ground truth, **c** segmentation result

M be the number of the one-to-one match pairs and N_1 and N_2 be the number of ground-truth and predicted text lines, respectively. The DR, RA, and FM metrics are calculated as follows:

$$DR = \frac{M}{N_1} \tag{8}$$

$$RA = \frac{M}{N_2} \tag{9}$$

$$FM = \frac{2 \times DR \times RA}{DR + RA} \tag{10}$$

4 Results and discussion

In our work, the text line is expressed with a mask to cover all the components in the text lines. We've trained three different GAN models concerning three loss definitions. These GAN models were implemented to different test datasets, obtained from VML-MOC, and VML-AHTE datasets. So, we've analyzed the segmentation performance of the models for different losses. GAN models can create noisy components in generated mask images in each training scenario. To clean up these noisy components, components smaller than a certain pixel size in the network outputs are deleted by the connected component analysis. Qualitative and quantitative results for each dataset are given in the following subsections.

4.1 Test results on VML-AHTE

As stated before, the VML-AHTE dataset contains challenges such as straight text lines with diacritics, and touching and overlapping characters between adjacent text lines. So, using this dataset makes it possible to determine the segmentation performance of the proposed GAN model under these challenges. Figure 7 implies that the GAN model mostly generates masks in accordance with the diacritics and touching/overlapping characters. As a comparison, segmentation results of the GAN model are given and compared with the ones of state-of-the-art unsupervised and supervised learning methods in Table 1. According to the numerical results, GAN models show very competitive results with other state-of-the-art methods. For the DR, RA, and FM metrics, the GAN model with L1 loss has the best performance of proposed GAN models and second best performance among all methods. For the IoU metrics, the GAN model with original GAN Loss has the second-best performance for Line IU and third-best for the Pixel IU.

4.2 Test results on VML-MOC

VML-MOC dataset is composed of skewed, curved, and multi-directional text lines. These cases exist in many historical handwritten documents and it's important to cope with these challenges and improve the segmentation performance in those documents. As can be seen in Figs. 8, 9, and 10, the proposed GAN model can generally adapt and learn the skewed, curved, and multi-oriented text lines. Table 2 compares the proposed GAN models' segmentation results with those of state-of-the-art unsupervised and supervised learning methods. According to the results, GAN models with

Table 2 Performance metric values on the VML-MOC dataset

Method	Learning	DR	RA	FM	Line IU	Pixel IU
Barakat et al. [18]	Unsupervised	–	–	–	60.99	80.96
FCN + EM[4]	Supervised	26.45	17.73	31.09	25.04	48.71
FCN + EM + Aug [4]	Supervised	84.43	58.89	68.77	35.12	60.97
GAN (GANLoss)	Supervised	56.52	52.07	53.74	51.34	79.92
GAN (L1)	Supervised	77.36	80.63	78.92	73.56	88.34
GAN (L2)	Supervised	79.73	82.94	81.22	72.80	89.30

L1 and L2 loss have the best IoU performance. Also, these models have better RA and FM and competitive DR performances.

As shown in Fig. 8 successful text line segmentation has been achieved even if the text lines are in different directions in regions where the text lines are more regular and have spaces. However, different orientations within a single text line or components such as notes and ornaments added to the end of the line can reduce the segmentation accuracy. As can be seen in Fig. 9, the proposed GAN model has a good segmentation performance on regular text lines. However, as in Fig. 10, the GAN model generates a combined mask for the text lines with different orientations in a small text region, resulting in a decrease in segmentation accuracy.

The numerical results show that the proposed GAN model performs good segmentation performance on two different challenging datasets, VML-AHTE and VML-MOC. On the VML-AHTE dataset, including narrow text lines with heavy diacritics, the GAN models achieved a good overall segmentation accuracy and failed to generate separate masks where there are nested text lines with touching and overlapping characters. The proposed GAN model achieved the best segmentation results among all supervised and unsupervised methods on the VML-MOC dataset. It implies that the GAN model can learn and generate masks for heavily skewed and curved text lines. However, in the case of sudden changes in the orientation of neighboring text lines, the GAN model predicts a single text line behavior and generates a mask for these text lines.

5 Conclusion

In this work, a GAN-based text line segmentation method has been proposed. TLS problem was considered an image-to-image translation problem and the GAN model was trained to learn the spatial information of text lines within patch images by generating text line masks. The GAN model was trained with images from two challenging datasets, VML-AHTE and VML-MOC. The main reason for selecting these datasets was to generate a model that can segment the skewed/curved text lines that exhibit rich diacritics, and touching/overlapping characters. In our work, we've trained the GAN model with different loss functions; GAN Loss, L1, and L2 loss. The resulting GAN models were tested on test images of VML-AHTE and VML-MOC datasets.

The results show that the proposed GAN model is very effective for the segmentation of the text lines of challenging handwritten documents and has a very high potential to be developed for this crucial problem in handwritten and historical document analysis. Considering that most historical documents contain skewed and curved text lines, touching and overlapping characters, footnotes, and/or side notes added with different orientations to the original manuscripts over the years, GAN models seem to have the ability to overcome these expectations.

Even though the text line segmentation problem has been resolved in printed documents, it remains a challenging and unresolved issue for handwritten documents.TLS techniques represent one of the fundamental steps employed in document analysis systems and are typically utilized in the initial stages of these systems. Consequently, errors made at this stage tend to propagate, leading to more significant errors throughout the system. Therefore, resolving the text line segmentation problem with high accuracy will enhance the accuracy of document analysis systems.TLS techniques are utilized in various tasks such as optical character recognition, handwritten document analysis, transcription tasks, document image understanding, word spotting, writer identification, content-based image retrieval, analysis of medical records, historical population records, and court records, among many other processes involving the analysis of handwritten documents. Although the main motivation of this work is on text line segmentation in challenging handwritten Arabic documents, the proposed model can also be modified and used in document de-warping tasks [10] following additional pre-processing steps

The proposed method has achieved more successful results in the text line segmentation task on the VML-AHTE dataset compared to the images in the VML-MOC dataset. The text lines in the images in the VML-AHTE dataset exhibit a more regular arrangement, while those in the VML-MOC dataset are skewed, curved, multi-directional, and irregular. The proposed method may make incorrect segmentation in cases where there is excessive space between two words in a text line or sudden changes in the direction of the line. By analyzing the gaps between lines of text, regions with more gaps than normal can be identified and it is thought that segmentation success can be increased by applying a different segmentation technique or post-processing steps in these areas. Similarly, various filters can be used to detect the orientation of lines of text, thus improving the accuracy of line segmentation by applying a more precise segmentation technique in areas where line orientations change rapidly.

Future work will include text line segmentation on Ottoman archival documents, which pose similar challenges to the datasets used in this paper and are studied by historians and researchers from around the world. For this purpose, we aim to create a large dataset, overcome the shortcomings of our method with various post-processing techniques, and segment text lines in Ottoman archival documents. Moreover, we plan to develop custom GAN models and present a holistic method for text line segmentation in challenging historical documents.

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Author Contributions All authors contributed to the conception and design. İÖ prepared the training and test datasets. İÖ and AAD carried out the training and testing stages. UÖ performed the verification and analysis. İÖ and AAD prepared all the figures and the original draft. UÖ prepared the tables; reviewed and edited the manuscript. All authors read and approved the final manuscript.

Data availibility No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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